

MAHESHBOOPATHI S

AI Engineer | Gen AI Specialist

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SUMMARY

AI Engineer with 2+ years of experience deploying production Agentic AI systems, advanced RAG pipelines, and cloud LLM workflows on AWS and Azure. Cut enterprise manual processing by 60% using MCP-integrated multi-agent workflows and improved inference accuracy by 40% using custom platform evaluation layers. Built a production real-time voice sales agent via the OpenAI Realtime API with RAG, achieving 90%+ context-aware retrieval accuracy. Experienced across the full AI lifecycle, from raw embedding pipelines to real-time MLOps monitoring.

EXPERIENCE

AI Engineer

SetNext.ai — Bengaluru, India

Apr 2025 - Aug 2026

Agentic AI Workflow Automation System

- Architected an end-to-end multiagent system to execute complex enterprise workflows, reducing manual processing time by 60%.
- Implemented Model Context Protocol (MCP) servers that securely connect decoupled AI agents to enterprise tools and internal data layers.
- Integrated advanced Pinecone-backed RAG infrastructure with specialized embeddings, achieving 90%+ context-aware query accuracy.
- Unified production model management, deployment, and inference by leveraging Azure AI Foundry, Azure AI Services, and Azure Machine Learning.

AI-Powered Real-Time Voice Sales Agent

- Delivered a production-ready, real-time voice sales agent using the OpenAI Realtime API to enable low-latency, multi-turn dialogue.
- Engineered chunking and real-time vector indexing pipelines to guarantee accurate conversational grounding, achieving 90%+ context-aware accuracy in live production use.
- Orchestrated tool-based sales workflows and automated follow-ups through native Azure AI Service integrations.
- Established real-time KPI dashboards within live production environments to track user engagement, token usage, and latency.

LLM Fine-Tuning-as-a-Service Platform

- Launched a cloud-native platform for scalable model fine-tuning, evaluation, and secure enterprise deployment.
- Optimized open-source LLMs using parameter-efficient fine-tuning workflows, including LoRA, PEFT, and QLoRA.
- Exposed automated REST APIs for dataset validation, version control, and multimodel pipeline tracking on AWS SageMaker and Azure ML.
- Augmented inference pipelines with targeted evaluation layers, boosting end-to-end response accuracy by 40%.

Software Trainee Intern

Aura Harks Technology — Chennai, India

Sep 2024 - Feb 2025

ML-Based Crop Yield Prediction Web Application

- Developed an end-to-end predictive web application to analyze localized soil, weather, and agricultural metrics for regional farming.
- Trained, engineered, and evaluated SVM and Linear Regression models, achieving a stable 85% predictive accuracy on test datasets.
- Constructed an operational ML inference backend using Python and Flask, with MySQL managing structured environmental records.

SKILLS

Generative & Agentic AI: LLMs, RAG, Prompt Engineering, Embeddings, LangChain, Multiagent Systems, MCP Server Development

LLM Fine-Tuning: LoRA, PEFT, QLoRA, Instruction Tuning, Dataset Preparation

Speech & Voice AI: ASR, Whisper, OpenAI Realtime API

Languages: Python, JavaScript, SQL

Backend & Messaging: FastAPI, Redis

Deep Learning: PyTorch, Hugging Face Transformers

Cloud & AI Platforms: AWS SageMaker, Azure Machine Learning, Azure AI Foundry, Azure AI Services

Vector Databases: Pinecone, FAISS

MLOps & Tools: Docker, Git, CI/CD, MLflow

CERTIFICATIONS

Data Analytics with Python — NPTEL, IIT Roorkee

12-week program covering probability, hypothesis testing, regression, and clustering

EDUCATION

Master of Computer Applications (MCA)

Muthayammal Engineering College, Namakkal, Tamil Nadu | 2022 - 2024